

# Effect of Aluminum Content on Microstructure and Mechanical Properties of $\text{Al}_x\text{CoCrFeMo}_{0.5}\text{Ni}$ High-Entropy Alloys

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High-entropy alloys  $\text{Al}_x\text{CoCrFeMo}_{0.5}\text{Ni}$  with varied Al contents ( $x = 0, 0.5, 1.0, 1.5$ , and  $2.0$ ) have been designed based on the  $\text{Al}_x\text{CoCrCuFeNi}$  system to improve mechanical properties for room and elevated temperatures. They have been investigated for microstructure and mechanical properties. As the aluminum content increases, the as-cast structure evolves from face-centered cubic dendrite + minor  $\sigma$ -phase interdendrite at  $x = 0$  to B2 dendrite with body-centered cubic (bcc) precipitates + bcc interdendrite with B2 precipitates at  $x = 2.0$ . This confirms the strong bcc-forming tendency of Al. The room-temperature Vickers hardness starts from the lowest, HV 220, at  $x = 0$ , attains to the maximum, HV 720, at  $x = 1.0$ , and then decreases to HV 615 at  $x = 2.0$ . Compared with the base alloy system, the current alloy system has a superior combination of hardness and fracture toughness. In addition,  $\text{Al}_x\text{CoCrFeMo}_{0.5}\text{Ni}$  alloys except  $x = 0$  display a higher hot hardness level than those of Ni-based superalloys, including In 718 and In 718 H, up to 1273 K and show great potential in high-temperature applications.

## INTRODUCTION

Most practical alloy systems are based on one principal metal element.<sup>1,2</sup> In 1995, Yeh proposed a novel concept utilizing high mixing entropy to design alloys.<sup>3,4</sup> By definition, a high-entropy alloy (HEA) contains at least five major metal elements, each of which has an atomic percentage between 5 at.% and 35 at.%. Several HEAs have been reported to possess useful properties such as high hardness, large work-hardening capacity, excellent high-temperature strength, superior temper-softening resistance, and good resistance to wear, oxidation, and corrosion.<sup>3–12</sup>

$\text{Al}_x\text{CoCrCuFeNi}$  HEAs ( $x = 0$  to  $3$ ) are interesting alloy systems that provide a wide range of microstructures, from face-centered cubic (fcc) to body-centered cubic (bcc), and strengths ranging from 385 MPa to 1617 MPa. These alloys are good candidates for structural components and tools. In addition, Cu in these alloys gives them excellent antibacterial properties. The effect of Al content on their hardness, crystal structure, high-temperature

strength, microstructure, and phase diagram has been studied.<sup>8,9</sup> However, because Cu has a positive mixing enthalpy with other elements in this alloy system, it tends to form a Cu-rich segregated phase in the interdendrite. As a result, an early incipient melting point of around 1000°C limits the application temperature of this alloy system. Moreover, the softness of the Cu-rich phase weakens these high-strength alloys and acts as a cathodic phase relative to other Cu-depleted phases, thereby reducing the corrosion resistance of the material and giving rise to pitting. Replacing Cu with Mo helps to overcome these drawbacks by generating a new alloy  $\text{Al}_x\text{CoCrFeMo}_{0.5}\text{Ni}$ . Previous researchers have varied the Fe,<sup>13</sup> Co,<sup>14</sup> Mo,<sup>15</sup> Cr,<sup>16</sup> and Ni<sup>17</sup> contents of this system to elucidate the individual effects of each element on the alloy microstructure and properties, which are typically superior to those of the base alloy system  $\text{Al}_x\text{CoCrCuFeNi}$ .

In this study, the effects of Al content on the microstructure and mechanical properties of  $\text{Al}_x\text{CoCrFeMo}_{0.5}\text{Ni}$  alloys (hereafter designated as Al- $x$ ) are investigated and compared with those of

the base alloy to characterize the effect of the replacement of Cu with Mo. The variation in the hot hardness of all Al- $x$  alloys is also investigated to illustrate their potential in high-temperature applications. This study contributes toward an overall understanding of Al-Co-Cr-Fe-Mo-Ni alloy systems, providing guidance for further alloy design.

## EXPERIMENTAL PROCEDURES

Conventional arc melting and casting under argon (which acted as a protective environment) were used to melt the Al, Co, Cr, Fe, and Ni granules with purity higher than 99 mass%, as well as the master alloy Fe-61.5 mass% Mo. Small round ingots with the desired alloy compositions were obtained. The microstructure and composition of these alloys were analyzed using a scanning electron microscope (SEM; JEOL JSM-5410; JEOL Ltd, Tokyo, Japan) with an attached x-ray energy dispersive spectrometer (SEM-EDS). The crystal structures of alloys were determined by an x-ray diffractometer (XRD; Rigaku ME510-FM2; Rigaku Corporation, Tokyo, Japan) using a copper target with an applied voltage of 30 kV and current of 20 mA, over a scan

range of  $20^\circ$  to  $100^\circ$  at a scanning rate of  $4^\circ/\text{min}$ . The Vickers hardness of the alloys was tested using a hardness testing machine (Mitutoyo HM-115; Mitutoyo America Corporation, Aurora, IL, USA) with a load of 5 kg for a duration of 15 s. Five measurements were taken for an average for each data point. Hot hardness was measured on a high-temperature hardness tester (Akashi AVK-CO) equipped with a heater and a vacuum chamber (Akashi AVK-HF), with the same load of 5 kg and duration of 15 s as used in measuring the room-temperature hardness. The specimens were 1.5 mm in thickness and 10 mm in diameter. Before measuring, the chamber was subjected to three cycles of purging with argon, was evacuated to  $1.9 \times 10^{-4}$  Torr, and finally filled with argon to approximately 1 atm pressure. In addition to the measurement at room temperature, the sample was measured at various temperatures with 100 K interval from 573 K to 1273 K. The heating rate was 10 K/min for each temperature increase.

## RESULTS AND DISCUSSION

### Constituent Phases and Microstructure

Figure 1 presents the XRD patterns of  $\text{Al}_x\text{CoCrFeMo}_{0.5}\text{Ni}$  as-cast alloys with various Al content. The crystal structure of these alloys comprises three simple phases: fcc, bcc, and  $\sigma$  phase. Their lattice parameters are summarized in Table I. The structure of the  $\sigma$  phase is similar to that of the CoCr phase, which is body-centered tetragonal with lattice constants  $a = 8.81 \text{ \AA}$  and  $c = 4.56 \text{ \AA}$ . Similar structures with slightly different lattice constants are also found in  $\sigma\text{-FeCr}$  and  $\sigma\text{-FeMo}$  phases, according to the JCPDS card. Related binary and ternary phase diagrams<sup>18</sup> show that the  $\sigma$  phase consists of Co, Cr, Fe, Ni, and Mo, in which Co, Fe, and Ni occupy the same lattice sites, while Cr and Mo occupy the other sites in the crystal structure.<sup>17</sup>

Al-0 and Al-0.5 alloys have two phases: fcc ( $a = 3.61 \text{ \AA}$ ) and  $\sigma$  phase ( $a = 8.92 \text{ \AA}$  and  $c = 4.61 \text{ \AA}$ ). The relative peak intensity of the  $\sigma$  and fcc phases shows that the volume fraction of the former is higher in the Al-0.5 alloy than in the Al-0 alloy.<sup>19</sup> The fcc phase is completely absent in the Al-1.0 alloy, which instead exhibits the  $\text{bcc}_1$  phase ( $a = 2.87 \text{ \AA}$ ). The ordered peak at  $31^\circ$  corresponds to the  $\text{bcc}_1$  phase; a previous TEM study on  $\text{Al}_x\text{CoCrFeMo}_{0.5}\text{Ni}$  identified the structure of this

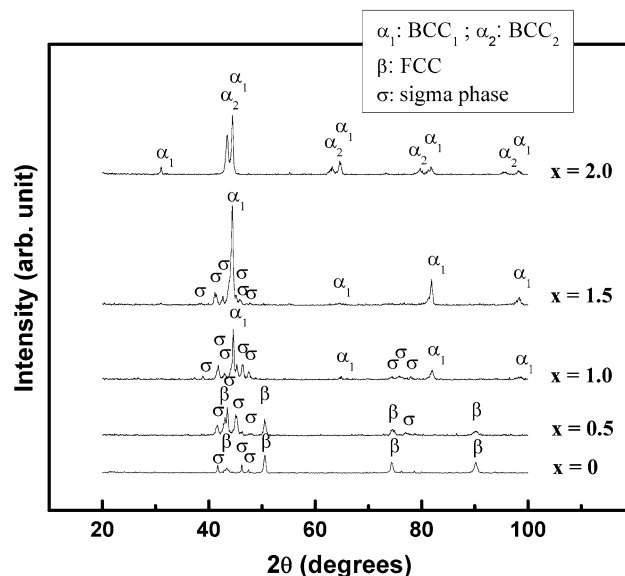


Fig. 1. XRD patterns of  $\text{Al}_x\text{CoCrFeMo}_{0.5}\text{Ni}$  alloys with various Al contents

Table I. Lattice parameters of constituent phases of  $\text{Al}_x\text{CoCrFeMo}_{0.5}\text{Ni}$  alloys

| Alloy  | BCC <sub>1</sub> (Å) | BCC <sub>2</sub> (Å) | FCC (Å) | BCT( $\sigma$ ) $a$ (Å) | BCT( $\sigma$ ) $c$ (Å) |
|--------|----------------------|----------------------|---------|-------------------------|-------------------------|
| Al-0   | —                    | —                    | 3.608   | 8.925                   | 4.599                   |
| Al-0.5 | —                    | —                    | 3.604   | 8.905                   | 4.624                   |
| Al-1.0 | 2.869                | —                    | —       | 8.909                   | 4.627                   |
| Al-1.5 | 2.879                | —                    | —       | 8.965                   | 4.642                   |
| Al-2.0 | 2.882                | 2.941                | —       | —                       | —                       |

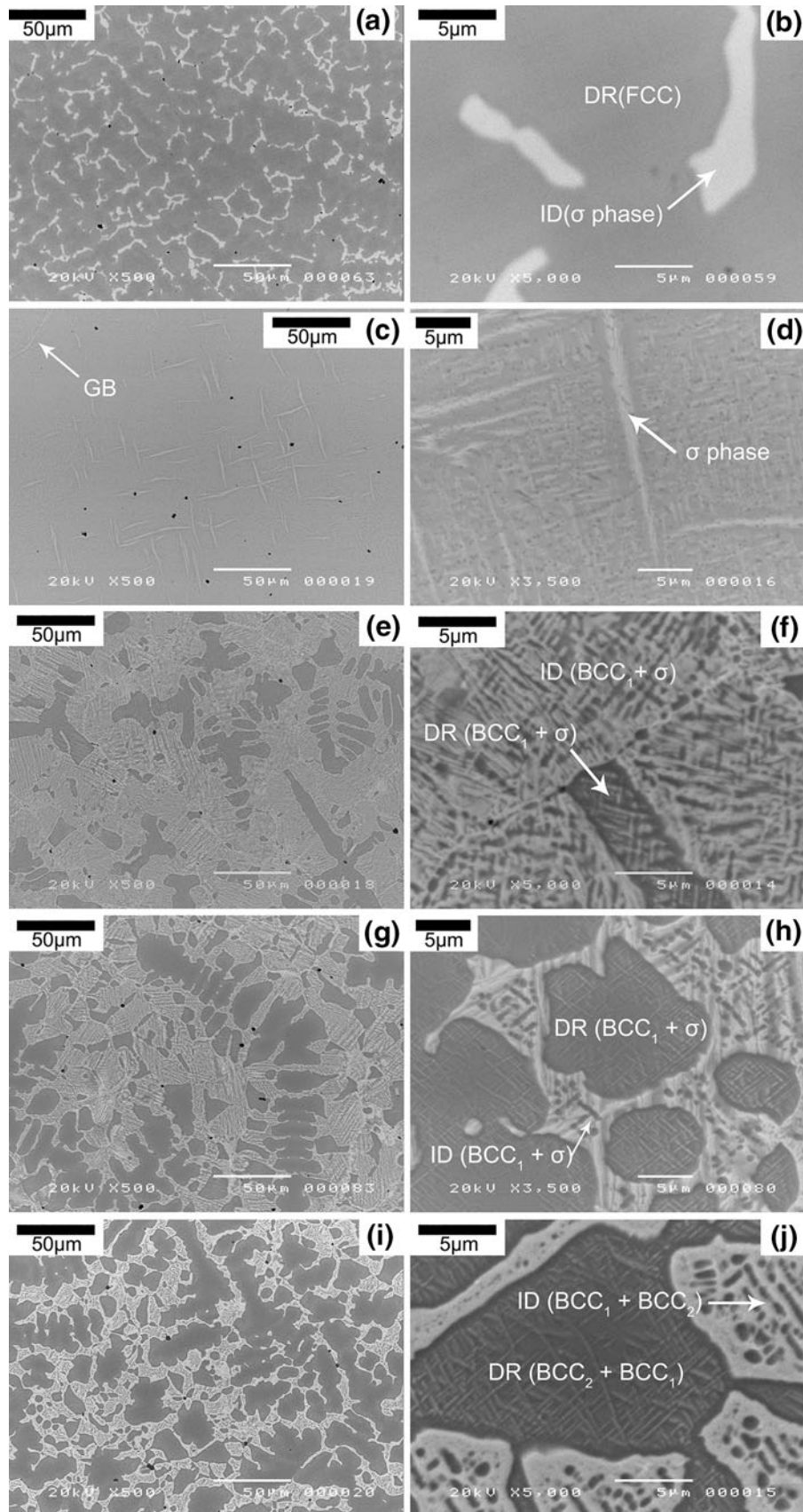


Fig. 2. Backscatter electron images of  $\text{Al}_x\text{CoCrFeMo}_{0.5}\text{Ni}$  alloys: (a) and (b) for  $x = 0$ , (c) and (d) for  $x = 0.5$ , (e) and (f) for  $x = 1.0$ , (g) and (h) for  $x = 1.5$ , and (i) and (j) for  $x = 2.0$

phase as B2.<sup>17</sup> Monitoring the relative peak intensity of the  $\sigma$  and  $\text{bcc}_1$  phases shows that the volume fraction of the  $\sigma$  phase decreases as the Al content increases from  $x = 1.0$  to  $1.5$ . The  $\sigma$  phase is completely absent in the Al-2.0 alloy, which is a mixture of  $\text{bcc}_1$  ( $a = 2.88 \text{ \AA}$ ) and  $\text{bcc}_2$  ( $a = 2.94 \text{ \AA}$ ). This result is in agreement with previous reports<sup>9</sup> stating that Al is an effective bcc phase stabilizer because of its strong  $p$ - $d$  bonding with transition metals even though it is an fcc element. In addition, it suggests that large fractions of Al can inhibit the formation of  $\sigma$  phase.

Figure 2 shows the as-cast microstructure of  $\text{Al}_x\text{CoCrFeMo}_{0.5}\text{Ni}$  alloys with various Al contents observed with SEM-BEI. The Al-0 alloy has a typical dendritic structure; no fine structures are observed in the dendrite and interdendrite phases. The Al-0.5 alloy displays a polycrystalline structure in which both large stringers around  $30 \mu\text{m}$  in length and fine stringers smaller than  $5 \mu\text{m}$  in the form of Widmanstätten pattern exist in grain interiors. In addition, another second phase also forms along grain boundaries. Because the grain size is quite large, approximately  $1000 \mu\text{m}$ , only a small segment of the grain boundary is shown in Fig. 2c. The Al-1.0, Al-1.5, and Al-2.0 alloys are also of dendritic structure, but both the dendrite and interdendrite further contain fine white phase. The volume fraction of the white phase is larger in the interdendrite phase than in the dendrite phase.

Table II shows the composition of the various regions and phases in the alloys studied in this investigation. The dendrite phase in the Al-0 alloy is close to the nominal composition, while that of the

interdendrite phase is richer in Cr and Mo. Combining these results with our XRD analysis indicates that the dendrite is the fcc phase and the white interdendrite is the  $\sigma$  phase whose composition is approximately  $(\text{Fe, Co, Ni})_{50}(\text{Cr, Mo})_{50}$ . The polycrystalline morphology of the Al-0.5 alloy suggests that its melting temperature range is small, and the Widmanstätten pattern suggests that a  $\sigma$  phase, with an approximate composition of  $(\text{Ni, Co, Fe, Al})_{65}(\text{Cr, Mo})_{35}$ , precipitates from the fcc matrix when the sample cools.

Table II also indicates that the dendrites of Al-1.0 and Al-1.5 alloys are richer in Al and Ni, while the interdendrites are richer in Cr and Mo. Because only the B2 and  $\sigma$  phases are evident in the XRD pattern of this sample, both the dendrite and interdendrite must have the bcc phase initially but transform to  $\sigma + \text{B2}$  phases during the cooling stage after solidification. Furthermore, because both dendrite and interdendrite have the same bcc crystal structure, Cr- and Mo-rich bcc phase in the interdendrite is in fact the outer layer of the Al- and Ni-rich cored dendrite; this layer is formed because of the partitioning of Cr and Mo during solidification.

The presence of substantial Co, Cr, Fe, and Mo in the Al- and Ni-rich bcc phase in the dendrite indicates that this phase is in fact an intermediate phase (i.e., a solid-solution compound) of the binary NiAl compound. The higher volume fraction of the white  $\sigma$  phase, which forms in the interdendrite during cooling, occurs because the fractions of Cr and Mo are larger in the interdendrite than in the dendrite. This interpretation has been previously confirmed for the Al-1.0 alloy by the TEM analysis

**Table II. Chemical composition of different regions and phases in  $\text{Al}_x\text{CoCrFeMo}_{0.5}\text{Ni}$  alloys**

| $x$ -Value | Region or phase         | Al   | Cr   | Fe   | Co   | Ni   | Mo   |
|------------|-------------------------|------|------|------|------|------|------|
| 0          | Nominal                 | 0    | 22.2 | 22.2 | 22.2 | 22.2 | 11.2 |
|            | DR (gray phase)         | 0    | 21.6 | 24.6 | 21.6 | 22.0 | 10.2 |
|            | ID (white phase)        | 0    | 26.1 | 18.2 | 17.0 | 14.5 | 24.2 |
| 0.5        | Nominal                 | 10.0 | 20.0 | 20.0 | 20.0 | 20.0 | 10.0 |
|            | Matrix (overall)        | 10.2 | 19.7 | 19.9 | 19.8 | 19.4 | 11.0 |
|            | Large string-like phase | 11.2 | 19.3 | 17.6 | 18.0 | 17.8 | 16.1 |
| 1.0        | Nominal                 | 18.2 | 18.2 | 18.2 | 18.2 | 18.2 | 9.0  |
|            | DR (overall)            | 24.5 | 12.3 | 15.6 | 18.1 | 24.4 | 5.1  |
|            | ID (overall)            | 11.3 | 21.0 | 19.3 | 17.5 | 14.1 | 16.8 |
| 1.5        | Nominal                 | 25.0 | 16.7 | 16.7 | 16.7 | 16.7 | 8.2  |
|            | DR (overall)            | 35.6 | 9.2  | 11.5 | 17.4 | 23.1 | 3.2  |
|            | ID (overall)            | 19.5 | 21.1 | 19.8 | 15.1 | 11.6 | 12.9 |
| 2.0        | Nominal                 | 30.8 | 15.4 | 15.4 | 15.4 | 15.4 | 7.6  |
|            | DR (overall)            | 39.0 | 9.3  | 11.6 | 16.3 | 20.4 | 3.4  |
|            | ID (overall)            | 23.4 | 22.0 | 18.5 | 12.6 | 9.1  | 14.4 |

DR dendrite, IR interdendrite.

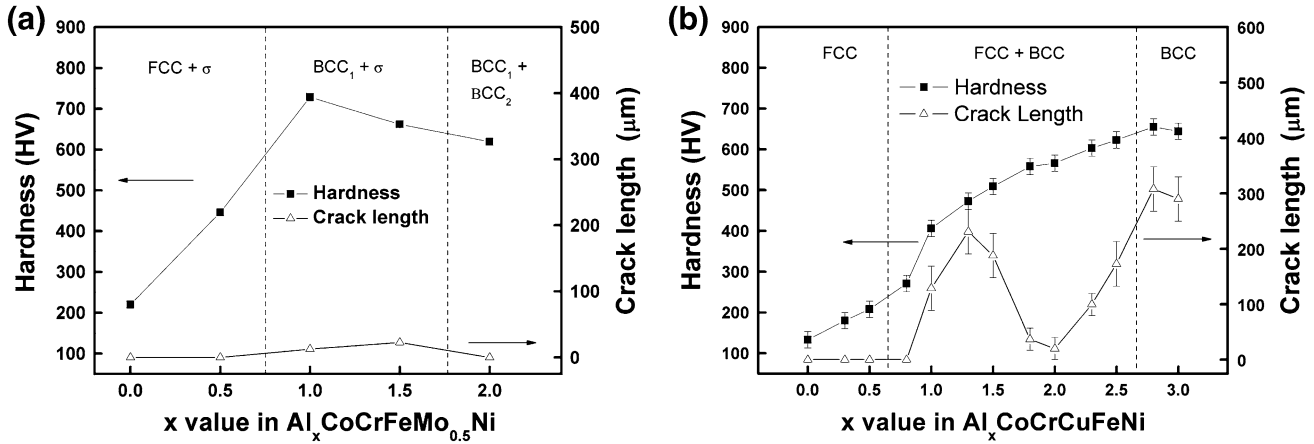


Fig. 3. (a) Vickers hardness and total crack length of  $\text{Al}_x\text{CoCrFeMo}_{0.5}\text{Ni}$  alloys as a function of Al content and (b) those of  $\text{Al}_x\text{CoCrCuFeNi}$  alloys for comparison with (a) (see Ref. 8).

of  $\text{Al}_x\text{CoCrFeMo}_{0.5}\text{Ni}$ .<sup>17</sup> Table II further shows that the dendrite phase of the Al-2.0 alloy is enriched in Al and Ni, while the interdendrite is enriched in Cr and Mo. Only two phases, a B2 phase and a disordered bcc ( $\text{bcc}_2$ ) phase, can be identified in the XRD pattern. The structure of each region in the SEM-BEI image can be inferred from the contrast: The white regions in Fig. 2(i) and (j) contain more heavy elements and they correspond to Cr- and Mo-rich disordered bcc phase. Thus, it is seen that the dendrite region contains  $\text{bcc}_2$  precipitates embedded in a B2 matrix. In contrast, the interdendrite region contains B2 precipitates embedded in a  $\text{bcc}_2$  matrix.

### Hardness and Crack Length

Figure 3 shows the hardness and crack length of  $\text{Al}_x\text{CoCrFeMo}_{0.5}\text{Ni}$  as a function of Al content. The corresponding values of the base alloy system  $\text{Al}_x\text{CoCrCuFeNi}$  are also compared with the current alloy system. The structural evolution, characterized in the Constituent Phases and Microstructure section, explains the differences in the hardness and crack length values between the samples. The large amount of soft fcc phase in the Al-0 alloy is responsible for its low Vickers hardness, HV 220 (Fig. 3a).<sup>20</sup> The larger amount of hard σ phase in Al-0.5 greatly increases its hardness to HV 440. The hardness further increases to HV 720 for the Al-1.0 alloy, resulting from the formation of a hard B2 phase at the expense of the fcc phase. However, the hardness of the Al-1.5 alloy is lower, HV 665, because of the reduction in the amount of σ phase that accompanies the increase in B2 phase. The hardness further declines to HV 615 for the Al-2.0 alloy because of the replacement of the hard σ phase with the softer  $\text{bcc}_2$  phase. To confirm that σ is the hardest phase, the Al-1.0 alloy was subjected to microhardness measurements on the dendrite (with less σ phase) and the interdendrite (with more σ phase). These measurements yield a hardness value of HV 634 for the dendrite and that of HV 824 for the interdendrite.

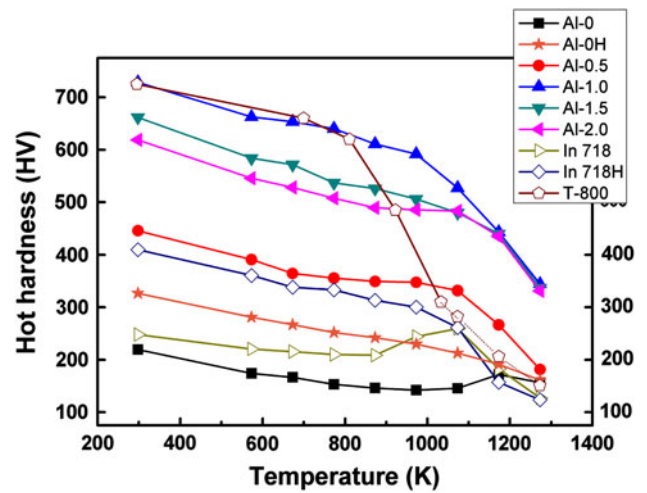


Fig. 4. Hot hardness ( $H$ ) as a function of temperature ( $T$ ) for  $\text{Al}_x\text{CoCrFeMo}_{0.5}\text{Ni}$  alloys, Ni-based superalloys of In 718 (as-received) and In 718 H (precipitation hardened), and Co-based superalloy of Triballoy T-800 (data are obtained from Stellite Company website).

As for the overall crack length (an indicator of fracture toughness<sup>21</sup>) variation, only Al-1.0 and 1.5 alloys exhibit a little cracking around the indentation marks. This suggests that the microstructure of the  $\text{Al}_x\text{CoCrFeMo}_{0.5}\text{Ni}$  alloys can prevent the formation and propagation of cracks to a large extent.

By comparing Fig. 3a with Fig. 3b, which shows measurements on the base alloys  $\text{Al}_x\text{CoCrCuFeNi}$  conducted under the same testing conditions, it can be seen that the hardness and toughness of the  $\text{Al}_x\text{CoCrFeMo}_{0.5}\text{Ni}$  alloys are superior to those of the base alloys  $\text{Al}_x\text{CoCrCuFeNi}$ . The improvement in the hardness is clearly attributable to the high volume fraction of the σ phase, which is harder than the bcc phase. The exact mechanism of the improvement in the toughness is not clear; however, its improvement even at a higher hardness level (for

example, at Al content of  $x = 1.0$  and  $1.5$ ) indicates that the mixture of B2 and  $\sigma$  phases in the current alloy is better than the mixture of fcc and bcc phases in the base alloy, even though the bcc phase usually decomposes into a spinodal modulated structure comprising bcc + ordered bcc (B2). Because the crystal structures of B2 and  $\sigma$  phases are very different from each other as analyzed in the Constituent Phases and Microstructure section, the fine mixture structure of the B2 and  $\sigma$  obviously provides many B2/ $\sigma$  incoherent phase boundaries to retard the crack propagation by crack deflection. On the other hand, bcc and B2 phases in the spinodal structure have the same lattice constant, and their coherent boundaries are expected to have little ability to deflect or retard the crack propagation.

### Hot Hardness at Elevated Temperatures

Figure 4 shows the hot hardness ( $H$ ) as a function of temperature ( $T$ ) for the  $\text{Al}_x\text{CoCrFeMo}_{0.5}\text{Ni}$  alloys,

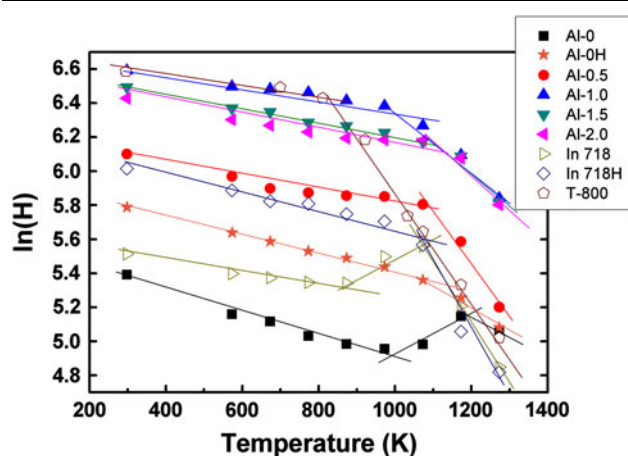


Fig. 5.  $\ln$  (hot hardness) plotted against temperature (from room temperature to 1273 K) for  $\text{Al}_x\text{CoCrFeMo}_{0.5}\text{Ni}$ , In 718, In 718 H, and T-800 alloys

as well as Ni-based superalloys of In 718 (as-received) and In 718H (precipitation-hardened), and Co-based superalloy of Tribaloy T-800 (data are obtained from Stellite Company website). Among the alloys shown in the plot, the T-800 alloy displays a much higher hardness at lower temperatures, but it softens rapidly above 800 K. The precipitation-hardened In 718H alloy has a medium-high hardness. The hardness of nonhardened In 718 is the lowest, but it increases above 900 K because of precipitation hardening. It is worth noting that these Ni-based alloys possess similar hardness values at 1100 K and soften at similar rates.

In contrast, Al- $x$  alloys exhibit a wide range of hardness levels depending on their Al content. Although they display a similar softening rate at lower temperatures, their resistance to softening is higher than that of the other alloys investigated in this section. Even though the Al-0 alloy has the lowest hardness level, it exhibits positive hardening at above 1000 K, and thus, it has a higher hardness than the In 718 and In 718H alloys. The hardening of the Al-0 alloy at above 1000 K is in fact related to its aging at high temperatures, similar to the behavior of In 718. To verify this, an age-hardening treatment at 1173 K for 2 h was applied to the Al-0 alloy, which was then tested for hot hardness. The age-hardened alloy, designated as Al-0H, has a higher hardness than the Al-0 alloy; however, it approaches the same hardness at 1273 K. The Al-1, Al-1.5, and Al-2.0 alloys can maintain the hardness levels at approximately HV 340 ( $\sim\text{Rc}35$ ) at 1273 K, higher than that of the In 718 and In 718 H alloys.

Figure 5 shows the data from Fig. 4 replotted as  $\ln(H)$  versus  $T$ . The experimental curve of each alloy fits two straight lines in stages I and II based on the Westbrook equation (see Eq. 1).<sup>17,20</sup> Related analyses and the physical meaning of each parameter can be found in the literature.<sup>17</sup> A change in the deformation mechanism occurs at the temperature where

Table III. Transition temperature ( $T_T$ ), intrinsic hardness ( $A$ ), and softening coefficient ( $B$ ) of  $\text{Al}_x\text{CoCrFeMo}_{0.5}\text{Ni}$  alloys and other superalloys

| Composition          | Transition temperature ( $T_T$ ) K | Below $T_T$ |                     | Above $T_T$ |                     |
|----------------------|------------------------------------|-------------|---------------------|-------------|---------------------|
|                      |                                    | $A_I$       | $B_I$               | $A_{II}$    | $B_{II}$            |
| Al-0                 | 1187                               | 260         | $-6.56 \text{ E-}4$ | 520         | $-0.95 \text{ E-}3$ |
| Al-0H                | 1062                               | 383         | $-5.36 \text{ E-}4$ | 955         | $-1.39 \text{ E-}3$ |
| Al-0.5               | 1088                               | 483         | $-3.66 \text{ E-}4$ | 8706        | $-3.02 \text{ E-}3$ |
| Al-1.0               | 995                                | 796         | $-2.99 \text{ E-}4$ | 3528        | $-1.82 \text{ E-}3$ |
| Al-1.5               | 1102                               | 758         | $-4.37 \text{ E-}4$ | 3300        | $-1.77 \text{ E-}3$ |
| Al-2.0               | 1100                               | 666         | $-3.29 \text{ E-}4$ | 3781        | $-1.89 \text{ E-}3$ |
| T-800 <sup>a</sup>   | 818                                | 791         | $-2.84 \text{ E-}4$ | 8041        | $-3.12 \text{ E-}3$ |
| In 718 <sup>b</sup>  | 1069                               | 268         | $-3.10 \text{ E-}4$ | 11766       | $-3.55 \text{ E-}3$ |
| In 718H <sup>c</sup> | 1037                               | 487         | $-5.29 \text{ E-}4$ | 13778       | $-3.74 \text{ E-}3$ |

<sup>a</sup>Composition of T-800: Co, 47.6%; Mo, 28%; Cr, 18%; Si, 3.4%; Fe, 1.5%; and Ni, 1.5%. <sup>b</sup>Composition of In 718: Ni, 53%; Cr, 19%; Fe, 18.5%; Nb, 5.1%; Mo, 3%; Ti: 0.9%; and Al, 0.5%. <sup>c</sup>Annealed at 1253 K for 30 min, air-cooled, and then aged at 1033 K for 8 h. This is then furnace cooled at a rate of 55 K, heated to 893 K for 8 h, and finally air cooled.

the two lines intersect. In the low-temperature region, the major deformation mechanism is slip, whereas in the high-temperature region, deformation is related to diffusional phenomena such as dislocation climb and grain boundary rotation.

Table III lists the values of transition temperature ( $T_T$ ), intrinsic hardness (A), and softening coefficient (B) calculated according to the Westbrook equation. The transition temperature of the Al- $x$  alloys ranges from 995 K to 1187 K. Except for the Al-1.0 alloy, all transition temperatures are approximately 300 K higher than that of T-800 and are slightly higher than that of In 718 and In 718 H. In addition, the current alloys in general have much lower softening coefficients  $B_{II}$ , and thus, much better softening resistance is observed in stage II. The reasons for these observations have been previously discussed<sup>17</sup> and are related mainly to the sluggish diffusion that is characteristic of HEAs; deformation in stage II is enhanced by vacancy diffusion through dislocation climb and grain boundary sliding mechanisms. Tsai et al.<sup>22</sup> verified the sluggish nature of HEA diffusion via diffusion couple experiments. Because the constituent phases, B2, bcc, fcc, and  $\sigma$ , of an  $\text{Al}_x\text{CoCrFeMo}_{0.5}\text{Ni}$  alloy system are multielement solid solutions,<sup>17</sup> these alloys should have a lower vacancy diffusion rate and thus better creep resistance than superalloys with a host element. Therefore, the current alloys show great promise for high-temperature applications that require higher strength and softening resistance.

## CONCLUSIONS

Based on the above discussion, the following conclusions can be drawn:

1. The addition of Al alters the crystal structure of the cast  $\text{Al}_x\text{CoCrFeMo}_{0.5}\text{Ni}$  alloy. At Al fraction  $x = 0$ , the structure comprises fcc dendrites and minor  $\sigma$ -phase interdendrites. At  $x = 0.5$ , the structure comprises fcc polycrystals and Widmanstätten  $\sigma$ -phase. At  $x = 1.0$  and 1.5, the structure comprises decomposed dendrites (B2 +  $\sigma$ -phase) and decomposed interdendrites (B2 +  $\sigma$ -phase), while at  $x = 2.0$ , the structure comprises decomposed dendrite (B2 + bcc) and decomposed interdendrite (B2 + bcc). This indicates that increasing the amount of Al promotes the formation of bcc and B2 phases but inhibits the formation of the  $\sigma$  phase.
2. The addition of Al greatly increases the hardness from HV 220 at  $x = 0$  to HV 440 at  $x = 0.5$  because of the increased amount of the hard  $\sigma$  phase. The hardness further increases to HV 720 at  $x = 1.0$  resulting from the formation of the hard B2 phase at the expense of the fcc phase. The hardness then decreases to HV 665 at  $x = 1.5$  because of the reduction in the amount of the hard  $\sigma$  phase that accompanies an increase in the amount of the bcc phase. The hardness further declines to HV 615 at  $x = 2.0$  as the amount of

the bcc phase further increases and replaces the hard  $\sigma$  phase.

3. The  $\text{Al}_x\text{CoCrFeMo}_{0.5}\text{Ni}$  alloys are harder and tougher than the base alloys  $\text{Al}_x\text{CoCrCuFeNi}$ . The large improvement in the hardness is attributable to the high volume fraction of the  $\sigma$  phase, which is harder than the bcc phase. The large improvement in the toughness at high hardness levels indicates that the mixture of B2 and  $\sigma$  phases performs better than the mixture of fcc and bcc phases that have a spinodal modulated B2 + bcc structure.
4. Tuning the Al content of the  $\text{Al}_x\text{CoCrFeMo}_{0.5}\text{Ni}$  alloy provides a wide range of hot hardness levels. Except for Al-1.0, all the alloys have transition temperatures around 300 K higher than that of the T-800 alloy and slightly higher than that of the In 718 and In 718 H alloys. The Al- $x$  alloys also in general have lower softening coefficients  $B_{II}$  and, thus, better softening resistance above their transition temperature, as compared to the In 718, In 718 H, and T-800 alloys. At 1273 K, the hardness of the Al-1.0, Al-1.5, and Al-2.0 alloys remains at around HV 340 ( $\sim\text{Rc35}$ ), which is much higher than that of the In 718 and In 718 H alloys. Therefore, the current alloys show great potential for high-temperature applications that require strong materials resistant to softening.
5. The excellent softening resistance of HEAs is mainly related to their sluggish diffusion. The low vacancy diffusion rate of the multielement solid-solution phases B2, bcc, fcc, and  $\sigma$  in the  $\text{Al}_x\text{CoCrFeMo}_{0.5}\text{Ni}$  alloys increase the creep resistance of these materials.

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